

Please note that phones and calculators are not allowed during the exam.

Exercise 1 (5.5 points). I) Five people are waiting for an elevator in a hotel. Each person is going to one of seven floors. How many ways are there for the five people to get off if:

1. Without any restrictions.
2. All five people get off on the same floor.
3. A and B get off on the same floor.
4. No two people get off on the same floor (all on different floors).
5. Two people get off on the same floor, and the other three each get off on different floors.

- II) 1. Using a combinatorial argument, show that $\binom{2n}{3} = 2\binom{n}{3} + 2n\binom{n}{2}$.
2. What is the coefficients of $x^{21}y^{135}z^{63}t^{17}v^{171}$ in the expansion of $(3x^7 - y^{27} - 7z^7 + t - 21v^{19})^{43}$?

Solution:

- I) 1. Each of the 5 people can choose any of the 7 floors independently. Therefore, the total number of ways is:

$$7^5. \quad (0.75pts)$$

2. All 5 people must choose the same floor. There are 7 possible floors, so the number of ways is:

$$7. \quad (0.5pts)$$

3. A and B must choose the same floor. There are 7 possible floors for them. The remaining 3 people can choose any of the 7 floors independently.

Thus, the number of ways is:

$$7 \cdot 7^3. \quad (0.75pts)$$

4. This is equivalent to assigning each of the 5 people to a unique floor from the 7 available floors. The number of ways is:

$$P(7, 5) = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3. \quad (0.75pts)$$

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5. First we choose the floor for the pair, there are 7 possible floors for the pair.

Choose the two people who will get off on that floor: There are $\binom{5}{2} = 10$ ways to choose the pair.

Assign the remaining 3 people to 3 different floors: The remaining 3 people must each choose a unique floor from the remaining 6 floors. The number of ways to assign them is $P(6, 3) = 6 \cdot 5 \cdot 4 = 120$.

Total number of ways:

$$7 \cdot \binom{5}{2} \cdot P(7, 3) = 7 \cdot \binom{5}{2} \cdot 7 \cdot 6 \cdot 5. \quad (1pts)$$

- II) 1. The left-hand side, $\binom{2n}{3}$, counts the number of ways to choose 3 elements from a set of $2n$ elements. Suppose the $2n$ elements are divided into two groups of size n each, Group A and Group B. There are four ways to choose 3 elements:

1. All 3 elements are from Group A, counted by $\binom{n}{3}$.

2. All 3 elements are from Group B, also counted by $\binom{n}{3}$.

3. 1 element is from Group A and 2 elements are from Group B, counted by $\binom{n}{1} \cdot \binom{n}{2}$.

4. 2 elements are from Group A and 1 element is from Group B, counted by $\binom{n}{2} \cdot \binom{n}{1}$.

The total number of ways to choose 3 elements from $2n$ elements is the sum of these cases:

$$2\binom{n}{3} + 2n\binom{n}{2}. \quad (1pt)$$

2. The multinomial expansion of $(3x^7 - y^{27} - 7z^7 + t - 21v^{19})^{43}$ is given by:

$$\sum_{k_1+k_2+k_3+k_4+k_5=43} \binom{43}{k_1, k_2, k_3, k_4, k_5} (3x^7)^{k_1} (-y^{27})^{k_2} (-7z^7)^{k_3} t^{k_4} (-21v^{19})^{k_5}.$$

The coefficient is:

$$\binom{43}{k_1, k_2, k_3, k_4, k_5} (3x^7)^{k_1} (-y^{27})^{k_2} (-7z^7)^{k_3} t^{k_4} (-21v^{19})^{k_5},$$

where $k_1 + k_2 + k_3 + k_4 + k_5 = 43$. We want to find the coefficient of $x^{21}y^{135}z^{63}t^{17}v^{171}$, such that

$$7k_1 = 21, \quad 27k_2 = 135, \quad 7k_3 = 63, \quad k_4 = 17, \quad 19k_5 = 171.$$

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Thus, $k_1 = 3$, $k_2 = 5$, $k_3 = 9$, $k_4 = 17$, and $k_5 = 9$.

Substituting these values into the multinomial expansion, we get:

$$\binom{43}{3, 5, 9, 17, 9} (3x^7)^3 (-y^{27})^5 (-7z^7)^9 t^{17} (-21v^{19})^9.$$

The coefficient of this term is:

$$\binom{43}{3, 5, 9, 17, 9} \cdot 3^3 \cdot (-1)^5 \cdot (-7)^9 \cdot 1^{17} \cdot (-21)^9. \quad (0.75pts)$$

Exercise 2 (3.5 points).

1. Prove that if 33 arbitrary words of any length are formed using distinct English letters, there must always be at least two words that share the same set of vowels.
2. Now, consider the list of the first occurrences of vowels. How many words must we form to ensure that at least two words contain the same list of vowels in the same order of their first occurrences?

For example, consider the words *education*, *equation*, and *automobile*. The words *education* and *equation* have the same list of first occurrences of vowels (e, u, a, i, o) in the same order. However, *automobile* contains the same vowels but in a different order: (a, u, o, i, e).

Note: The English vowels are {a, e, i, o, u}.

Solution:

1.
 - **Pigeons:** The 33 words. (0.25pt)
 - **Pigeonholes:** All possible subsets of vowels {a, e, i, o, u}.
 - Total subsets, each word contain a set 0, 1, 2, 3, 4 or 5 vowels so

$$\underbrace{\binom{5}{0}}_{0 \text{ vwls}} + \underbrace{\binom{5}{1}}_{1 \text{ vwls}} + \underbrace{\binom{5}{2}}_{2 \text{ vwls}} + \underbrace{\binom{5}{3}}_{3 \text{ vwls}} + \underbrace{\binom{5}{4}}_{4 \text{ vwls}} + \underbrace{\binom{5}{5}}_{5 \text{ vwls}} = \sum_{k=0}^5 \binom{5}{k} = 2^5 = 32. \quad (0.75pt)$$

- By the pigeonhole principle: $33 > 32 \implies$ At least two words share the same subset of vowels. (0.5pt)
2. Now we seek to find number of words, so we seek to find how many pigeons
 - **Pigeonholes:** Distinct ordered lists of first vowel appearances.
 - For k vowels ($0 \leq k \leq 5$), the number of ordered lists is the falling factorial:

$$\binom{5}{k} k! = (5)_k = 5 \cdot 4 \cdot \dots \cdot (5 - k + 1).$$

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– Total pigeonholes:

$$\underbrace{(5)_0}_{0 \text{ ordrd vwls}} + \underbrace{(5)_1}_{1 \text{ ordrd vwls}} + \underbrace{(5)_2}_{2 \text{ ordrd vwls}} + \underbrace{(5)_3}_{3 \text{ ordrd vwls}} + \underbrace{(5)_4}_{4 \text{ ordrd vwls}} + \underbrace{(5)_5}_{5 \text{ ordrd vwls}} = \sum_{k=0}^5 (5)_k$$

$$= 1 + 5 + 20 + 60 + 120 + 120 = 326. \quad (1pt)$$

– **Pigeons:** We need to form at least 327 words. (0.5pt)

- By the pigeonhole principle: $327 > 326 \implies$ At least two words share the same ordered list. (0.5pt)

Exercise 3 (4 points). Consider $2n$ elements that belong to n different kinds, with two like elements of each kind, and let V_n be the number of permutations of these elements in which no two like elements are consecutive. Find V_n .

Solution:

Let Ω be the number of all perumutations of the $2n$ elements, then $|\Omega| = \frac{(2n)!}{2^n}$. (0.75pts) and let A_i be the set of permutations where two elements of the i -th kind are consecutive. We want to find the number of permutations where none of the A_i occur (0.5pts). We want to find $|\overline{A_1} \cap \overline{A_2} \cap \dots \cap \overline{A_n}|$. (0.25pts)

$|A_1|, |A_2|, \dots, |A_n|$ are the number of permutations where two elements of the i -th kind are consecutive. We can calculate these as follows:

$$|A_1| = \frac{(2n-1)!}{2^{n-1}}, |A_2| = \frac{(2n-1)!}{2^{n-1}}, \dots, |A_i| = \frac{(2n-1)!}{2^{n-1}}.$$

Then,

$$\sum_{i=1}^n |A_i| = n \cdot \frac{(2n-1)!}{2^{n-1}}.$$

$|A_1 \cap A_2|, |A_1 \cap A_3|, \dots, |A_i \cap A_j|$ are the number of permutations where two elements of the i -th and j -th kind are consecutive. We can calculate these as follows:

$$|A_1 \cap A_2| = \frac{(2n-2)!}{2^{n-2}}, |A_1 \cap A_3| = \frac{(2n-2)!}{2^{n-2}}, \dots, |A_i \cap A_j| = \frac{(2n-2)!}{2^{n-2}}.$$

Then,

$$\sum_{1 \leq i < j \leq n} |A_i \cap A_j| = \binom{n}{2} \cdot \frac{(2n-2)!}{2^{n-2}}.$$

For $|A_1 \cap A_2 \cap \dots \cap A_k|$, we have:

$$|A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}| = \frac{(2n-k)!}{2^{n-k}}.$$

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Then,

$$\sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} |A_1 \cap A_2 \cap \dots \cap A_k| = \binom{n}{k} \cdot \frac{(2n-k)!}{2^{n-k}}. \quad (1.25pts)$$

We have $V_n = |\Omega| - |\bigcup_{i=1}^n A_i|$. Using the inclusion-exclusion principle, we can write

$$\begin{aligned} V_n &= |\Omega| - \sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} |A_1 \cap A_2 \cap \dots \cap A_k| \\ &= \frac{(2n)!}{2^n} - \sum_{k=1}^n (-1)^{k-1} \binom{n}{k} \cdot \frac{(2n-k)!}{2^{n-k}} \\ &= \sum_{k=0}^n (-1)^{k-1} \binom{n}{k} \cdot \frac{(2n-k)!}{2^{n-k}}. \quad (1.25pts) \end{aligned}$$

Exercise 4 (4 points). How many integer solutions exist for the equation

$$(x_1 + x_2 + \dots + x_m)(y_1 + y_2 + \dots + y_n) = 25,$$

given that x_i are all of the same sign, and y_i are all of the same sign?

Solution:

- The integer 25 has 4 possible factorizations as a product of two integers:

$$25 = 25 \times 1 = 5 \times 5 = -1 \times -25 = -5 \times -5. \quad (0.5pt)$$

The positive factorizations corresponds to the following cases:

$$\begin{cases} (x_1 + x_2 + \dots + x_m) = 25 \text{ (or 1 resp.)} \\ x_i \geq 0 \quad \forall i \end{cases} \quad \text{and} \quad \begin{cases} (y_1 + y_2 + \dots + y_n) = 1 \text{ (or 25 resp.)} \\ y_j \geq 0 \quad \forall j \end{cases}$$

or

$$\begin{cases} (x_1 + x_2 + \dots + x_m) = 5 \\ x_i \geq 0 \quad \forall i \end{cases} \quad \text{and} \quad \begin{cases} (y_1 + y_2 + \dots + y_n) = 5 \\ y_j \geq 0 \quad \forall j \end{cases}$$

The same for the negative values corresponds to the following cases:

$$\begin{cases} (x_1 + x_2 + \dots + x_m) = -25 \text{ (or -1 resp.)} \\ x_i \leq 0 \quad \forall i \end{cases} \quad \text{and} \quad \begin{cases} (y_1 + y_2 + \dots + y_n) = -1 \text{ (or -25 resp.)} \\ y_j \leq 0 \quad \forall j \end{cases}$$

or

$$\begin{cases} (x_1 + x_2 + \dots + x_m) = -5 \\ x_i \leq 0 \quad \forall i \end{cases} \quad \text{and} \quad \begin{cases} (y_1 + y_2 + \dots + y_n) = -5 \\ y_j \leq 0 \quad \forall j \end{cases}$$

Identify Systems (1.5pts)

Final Exam Solution

For each positive case, the number of solutions is given by:

$$\binom{m+24}{25} \binom{n}{1} + \binom{m}{1} \binom{n+24}{25} + \binom{m+4}{5} \binom{n+4}{5}$$

Solve Systems (1.5pts)

and by symmetry the same solution of the negative case, so the total number of solution is

$$2 \left(\binom{m+24}{25} \binom{n}{1} + \binom{m}{1} \binom{n+24}{25} + \binom{m+4}{5} \binom{n+4}{5} \right) \text{ (0.5pt)}$$

First name: _____ Last name: _____ Group: _____

Exercise 5 (3 points). *Decide whether each of the following statements is true or false. Provide a justification for your answer using an example or a counterexample.*

1. *Every subgraph of a connected graph is also connected.*

.....

2. *A spanning subgraph of a graph G contains all the edges of G .*

.....

3. *A cycle graph C_n is a spanning subgraph of a complete graph K_n .*

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4. *Every edge-induced subgraph is also a vertex-induced subgraph.*

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5. *If two vertices in a graph share exactly one neighbor, they must be adjacent to each other.*

.....

6. *A graph with 10 vertices and no edges has 10 components.*

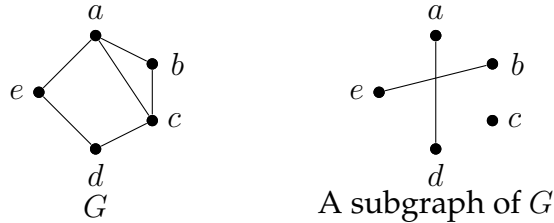
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Solution:

1. False.

A subgraph of a connected graph is not necessarily connected.

Counterexample:



2. False.

A spanning subgraph of G contains all the vertices of G , but not necessarily all the edges.

(0.5pts)

3. True.

A cycle graph C_n (where $n \geq 3$) is a subgraph of K_n that includes all n vertices. Since C_n includes all the vertices of K_n , it satisfies the definition of a spanning subgraph. (0.5pts)

4. False.

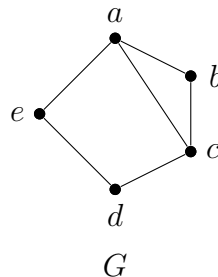
An edge-induced subgraph is formed by a subset of edges and their endpoints. However, a vertex-induced subgraph is formed by a subset of vertices and all edges between them.

These definitions are not equivalent. (0.5pts)

5. False.

Two vertices sharing exactly one common neighbor do not have to be adjacent. (0.5pts)

Counterexample:



The vertices e and c share exactly one neighbor, vertex d , but they are not adjacent. (0.5pts)

6. True.

If a graph has 10 vertices and no edges, each vertex forms its own connected component.

Therefore, the graph has exactly 10 components. (0.5pts)